

A Minor Project II report on

NeoNote

Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Engineering in Software Engineering at Pokhara University

By

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(May 04, 2024)

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Acknowledgement

We would like to extend our heartfelt gratitude to our project supervisor **Er. Tara Bahadur Thapa** for her invaluable guidance and continuous support throughout the development of NeoNote. We are equally thankful to the principal **Dr. Bishwo Udhir Poudel Gharti** and vice principal **Mr. Ashok Raj Parajuli** for providing us with this opportunity to apply our learning in a real-world project. Our sincere appreciation also goes to the entire team of supervisors who helped us overcome various challenges during the development process. A special thanks to our Project Head **Er Santosh Panth** for his constant motivation and leadership, which greatly inspired us throughout the project journey.

We are also immensely grateful to our friends and peers who stood by us during the past few months—sharing ideas, solving problems, and contributing their efforts to enhance the project. Their collaborative spirit and encouragement were crucial in shaping NeoNote into a fully functional application with note-taking, diary, task management, and collaboration features. The entire team is truly thankful to everyone who directly or indirectly contributed to the successful completion of this project.

Anil B.k

Kriti Adhikari

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ABSTRACT

NeoNote is a comprehensive platform independent app developed to address the need for efficient task management, note-taking, and project organization. The project aims to simplify the complexities of everyday planning by providing a unified platform that combines intuitive note-taking. Built using Flutter for the frontend, Django for the backend, and PostgreSQL for database management, the system ensures seamless functionality, scalability. The methodology involves creating an interactive interface that enables users to create, organize, and edit notes while managing tasks. Initial testing highlights smooth user interactions, reliable performance, and a responsive design. The system's effectiveness is evaluated through internal performance metrics and user trials, ensuring its suitability for both individual and collaborative use cases. **NeoNote** concludes by offering a versatile, efficient, and user-centric solution that empowers individuals and teams to optimize their productivity and achieve their goals.

Keywords: Platform Independent app, task management, note-taking, project organization, productivity, Flutter, Django, PostgreSQL, scalable, performance optimization.

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Chapter 1

INTRODUCTION

1.1 Background

NeoNote is a platform-independent productivity application designed to streamline task management, note-taking, and project organization across a wide range of devices and operating systems. The primary objective of NeoNote is to enhance user productivity by bringing together essential tools into a single, integrated platform that reduces the need for switching between multiple applications. With an intuitive and user-friendly interface, NeoNote is suitable for individuals, students, professionals, and teams who wish to improve their workflow and maintain better control over their tasks and projects. The application offers users the ability to create, categorize, and organize notes and tasks according to their personal preferences, whether by project, topic, deadline, or priority. Its flexible and modular design allows for a personalized workspace, enabling users to adapt the platform to their specific needs and working styles. NeoNote also features visual progress tracking, helping users stay focused on their goals by clearly displaying task statuses, deadlines, and overall progress. This visual feedback promotes better time management and motivation. In addition, the platform supports collaborative features, allowing multiple users to contribute to notes, tasks, or entire projects, making it suitable for both personal and team use. By offering a comprehensive set of productivity tools within a single application, NeoNote empowers users to stay organized, maintain clarity, manage their time effectively, and ultimately achieve their objectives more efficiently.

1.2 Problem statement

NeoNote is an all-in-one productivity tool that enables users to take notes, write diaries, set goals, manage tasks, and schedule events. It also supports team collaboration by allowing users to create groups and assign tasks to team members for efficient workflow management.

1.3 OBJECTIVE

The primary objective of our project is:

- The objective of this project is to build a user-friendly platform that allows users to manage notes, tasks, and projects efficiently, supports collaboration through user invitations and shared workspaces, sends real-time task notifications, and includes a time zone-aware calendar with access controls for event management.

1.4 IMPLICATION

NeoNote brings together essential productivity tools like note-taking, diary writing, task tracking, goal setting, and event scheduling into a single, user-friendly platform. By allowing users to organize their thoughts, plan effectively, and assign tasks within groups, it promotes better time management and teamwork. This helps individuals and small teams stay focused, organized, and aligned toward their goals without relying on multiple separate tools.

Chapter 2

LITERATURE REVIEW

Many productivity apps have already been developed that provide task management and note-taking services. Some of them include Google Keep, ClickUp, and Notion.

1.1 Google Keep

Google Keep is a lightweight, cross-device note-taking app that syncs seamlessly with your Google account. It lets you jot down quick notes, create checklists, and set time- or location-based reminders, all while organizing items with colors and labels. Voice memos can be recorded and transcribed automatically, making it perfect for capturing ideas on the fly and keeping your personal tasks in order.

1.2 ClickUp

ClickUp is an all-in-one productivity platform that adapts to any workflow, combining task management, goal tracking, time logging, and document creation. With multiple view options—List, Board, Calendar, and Gantt—teams can choose the layout that best suits their project style. Integrated comments, mentions, and hundreds of third-party integrations streamline collaboration, while advanced customization scales from small teams to enterprise needs.

1.3 Notion

Notion is a modular workspace where notes, tasks, databases, and wikis coexist in customizable pages made up of “blocks.” Its block-based design empowers users to mix text, tables, kanban boards, calendars, and more within a single document. Shared workspaces and real-time editing enable teams to collaborate on everything from meeting notes to complex project trackers, making it a versatile hub for both simple lists and comprehensive knowledge bases.

Chapter 3

TOOLS AND METHODOLOGY

3.1 REQUIRED TOOLS

Different tools used to develop our app:

- **Flutter:** Best for cross-platform deployment and creating visually rich, interactive apps.
- **Django:** For backend development, handling server-side logic and database management.
- **PostgreSQL:** For managing the system's relational database, storing user data, notes, and tasks.
- **Figma:** For designing and prototyping the app's user interfaces.
- **Draw.io:** For creating system architecture diagrams and workflows.
- **MS-Word:** For writing and preparing reports.

3.2 Designs

3.2.1 Use-Case Diagram

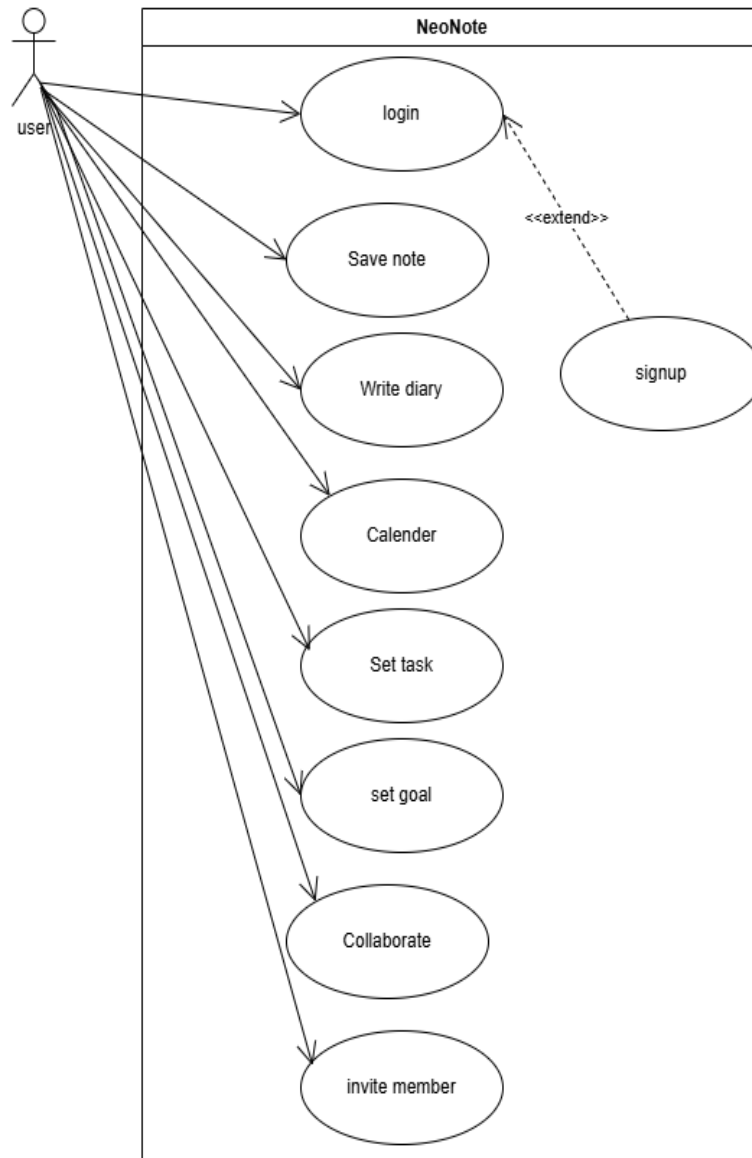


FIGURE 1: USE CASE DIAGRAM

USE CASE: NeoNote

Actors:

1. **User:** A person who uses NeoNote for note-taking, task management, and project organization.

Use Cases:

1. **Login:** Users and administrators can log in to the system using Firebase Authentication.
2. **Logout:** Users and administrators can log out of the system at any time.
3. **Create Note:** Users can create, edit, and save new notes.
4. **Organize Tasks:** Users can add, edit, and delete tasks related to their notes and projects.
5. **View Notes/Tasks:** Users can view their saved notes and tasks, organized by projects or categories.
6. **Set Reminders:** Users can set reminders for tasks or notes to receive notifications at a specified time.
7. **Share projects:** Users can share projects with other users for collaborative editing and viewing.
8. **Delete Note/Task:** Users can delete any note or task they no longer need.

3.2.2 Entity Relation Diagram (EDR)

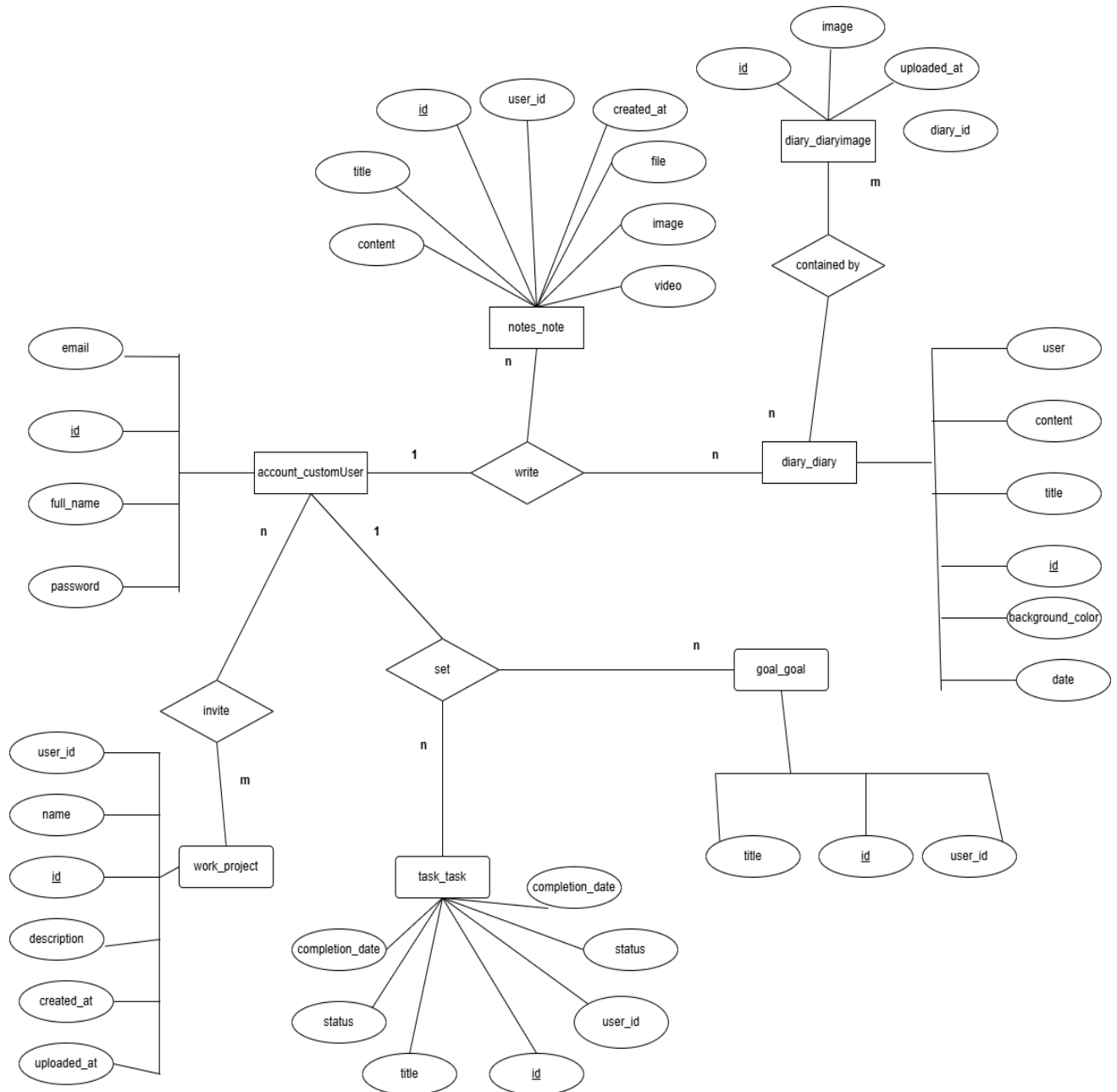


FIGURE 2: ENTITY RELATIONSHIP DIAGRAM

3.2.3 System Sequence Diagram (SSD)

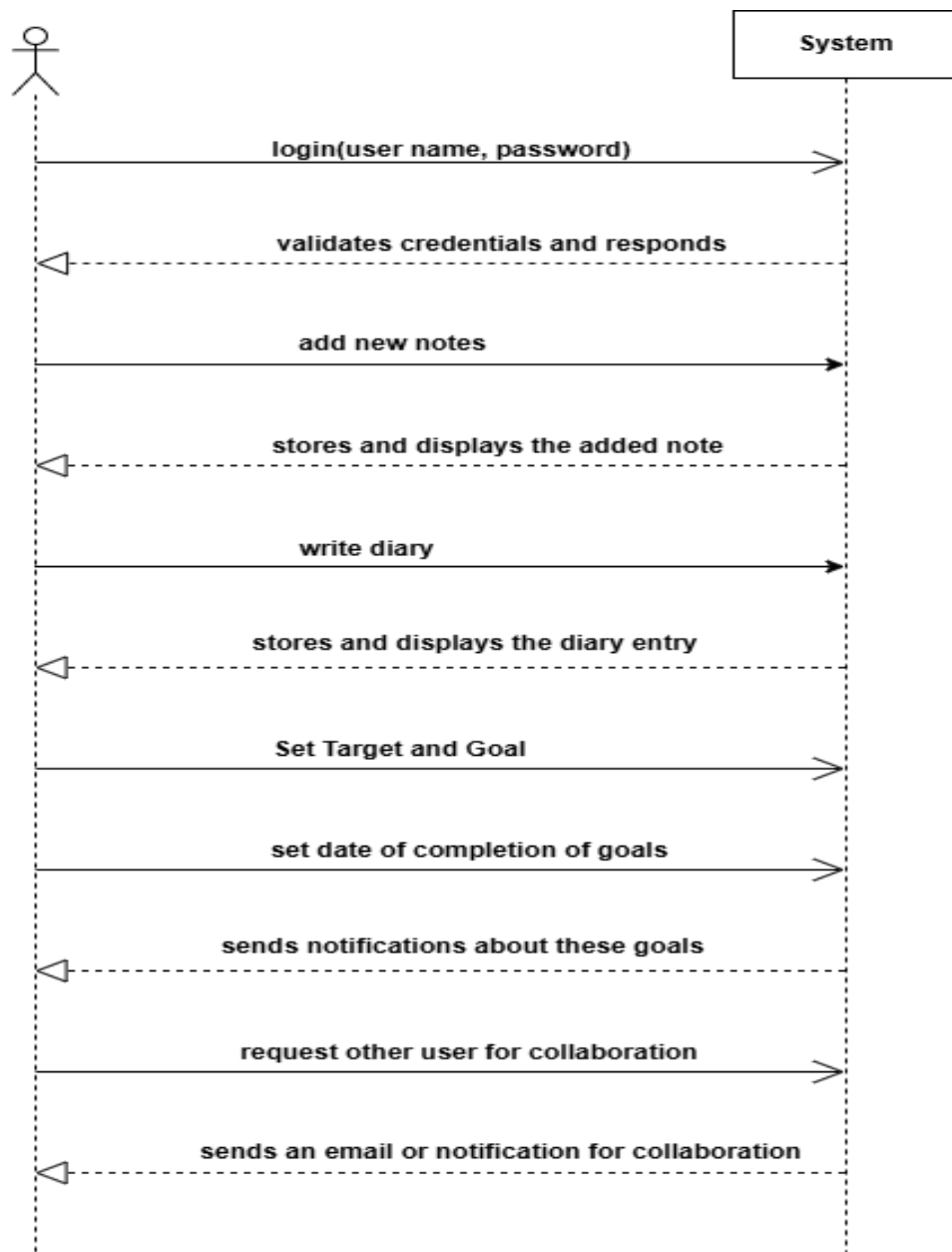


FIGURE 3: SYSTEM SEQUENCE DIAGRAM (SSD)

3.2.4 DESIGN CLASS DIAGRAM

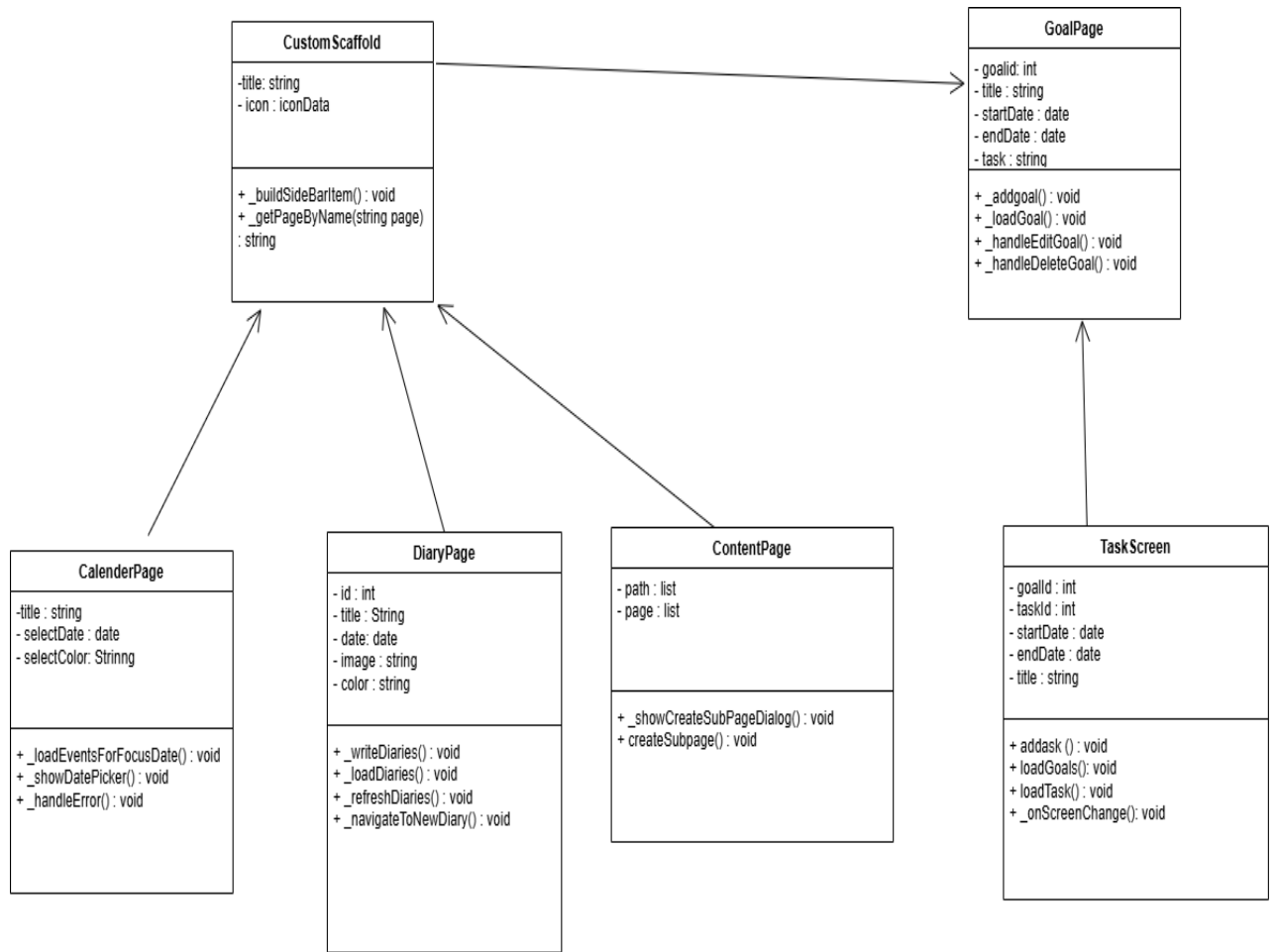


FIGURE 4: DESIGN CLASS DIAGRAM

3.2.5 INTERACTION DIAGRAM

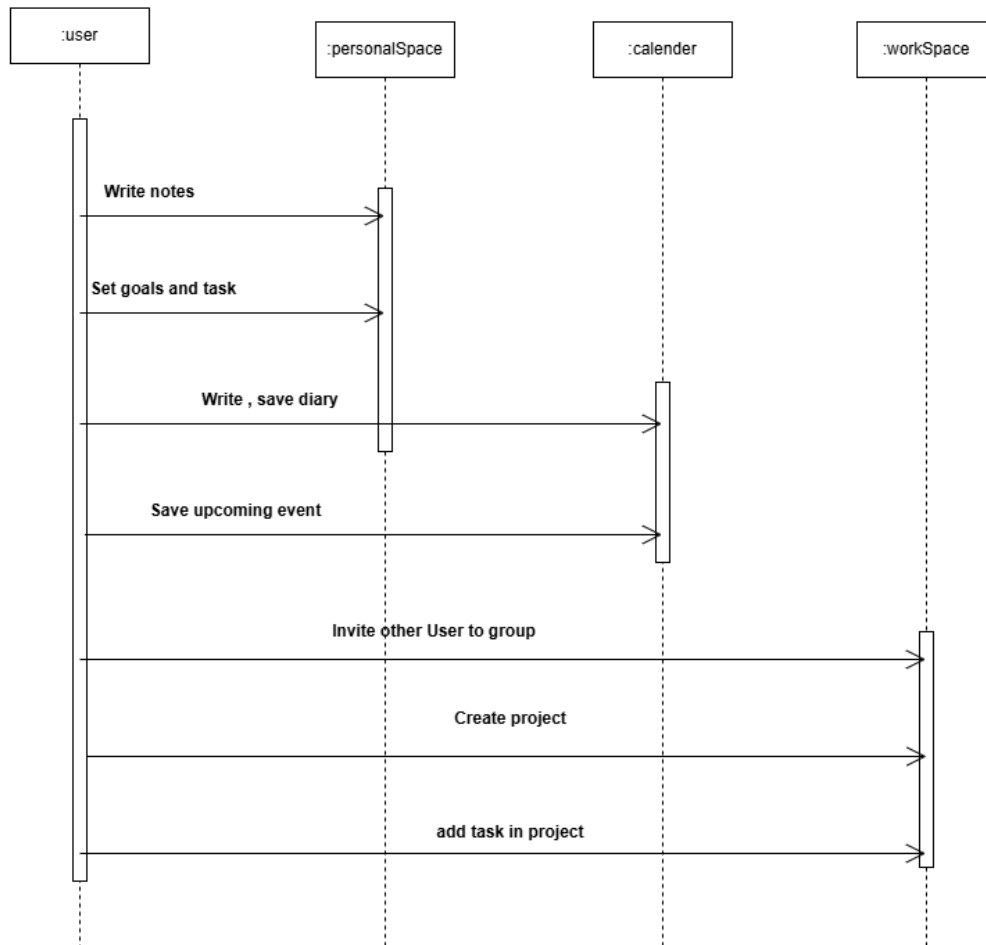


FIGURE 4: INTERACTION DIAGRAM

3.3 APPROACH

For the development of **NeoNote**, an **agile approach** is used to ensure flexibility and continuous improvement throughout the project. This iterative approach allows for frequent updates, quick feedback loops, and the ability to adapt to changing user needs. Each sprint will focus on specific features, such as task management, note-taking, and real-time collaboration, ensuring that the app evolves efficiently while maintaining a user-centric focus. This methodology also supports frequent testing and refinement to guarantee optimal performance and user satisfaction.

CHAPTER 4

Findings, Results and Analysis

4.1 Test Case Report/ Result

Table 1: TEST CASE

S.N	TEST CASE	EXPECTED	OBSERVED	RESULT
1	Authentication	Enter empty id and password	The id and password is empty.	Pass
2	Authentication	Enter incorrect password	Password is incorrect.	Pass
3	Check Invalid Password	Set password less than 8 character	The password is at least 8 characters.	Pass
4	Authentication	Enter delete id and password of left user	This id does not exist.	Pass
5	Ensure that users can create a new note successfully.	Create a New Note	The new note is saved and visible in the note list.	Pass
6	Ensure changes made by one user are reflected instantly for another user.	Collaboration (task Editing)	Users see the changes immediately.	Pass
7	Verify that users can create, edit, and delete tasks.	Task Management	The task is created, edited, and deleted successfully, with changes reflected in the app.	Pass

4.2 key output

1. It provides an interface to take and organize notes.
2. It includes a diary feature to record personal thoughts.
3. It allows users to collaborate and share notes.
4. It helps manage daily tasks and long-term goals.

4.3 ANALYSIS

Our project NeoNote is a note-taking and productivity application designed to help users manage personal diaries, tasks, goals, and collaborate through shared notes. It includes essential features like real-time note sharing, task tracking, and secure cloud storage. The app has undergone all required testing and produced the expected results. Working on NeoNote helped us gain practical experience in full-stack development, including frontend, backend, API development, and integration. The product is now ready for real-world use. We applied Black Box Testing to validate the application's functionality without accessing internal code. This approach focused on inputs, outputs, and expected behaviors based on user requirements.

Despite its strengths, NeoNote has a few limitations:

- Requires internet access to function.
- Manual support may be needed to maintain collaboration quality.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

Developing **NeoNote** gave us valuable experience in building a system that blends functionality, usability, and collaboration. From UI design to backend and API integration, we explored new areas and overcame challenges with the help of online resources and developer communities.

We also recognized the importance of testing and quality assurance, focusing on meeting user requirements through structured testing rather than manual checks. Tools like GitHub, Trello, and Discord played a key role in helping us collaborate effectively.

NeoNote was proposed to enhance personal productivity and note-sharing, and we are confident that we have successfully brought our vision to life as a practical and reliable application.

5.2 RECOMMENDATION

NeoNote has reached the end of its development timeline, and we've successfully met our initial goals. However, there is still room for improvement. Based on our experience, we recommend the following enhancements:

1. Implement cloud-based data storage and synchronization to ensure user data is automatically backed up, available across devices, and resilient to local failures.
2. Create a Favorites section that lets users bookmark and quickly access their most-used notes, tasks, and projects.
3. Add Templates functionality, allowing users to choose or create reusable templates for notes

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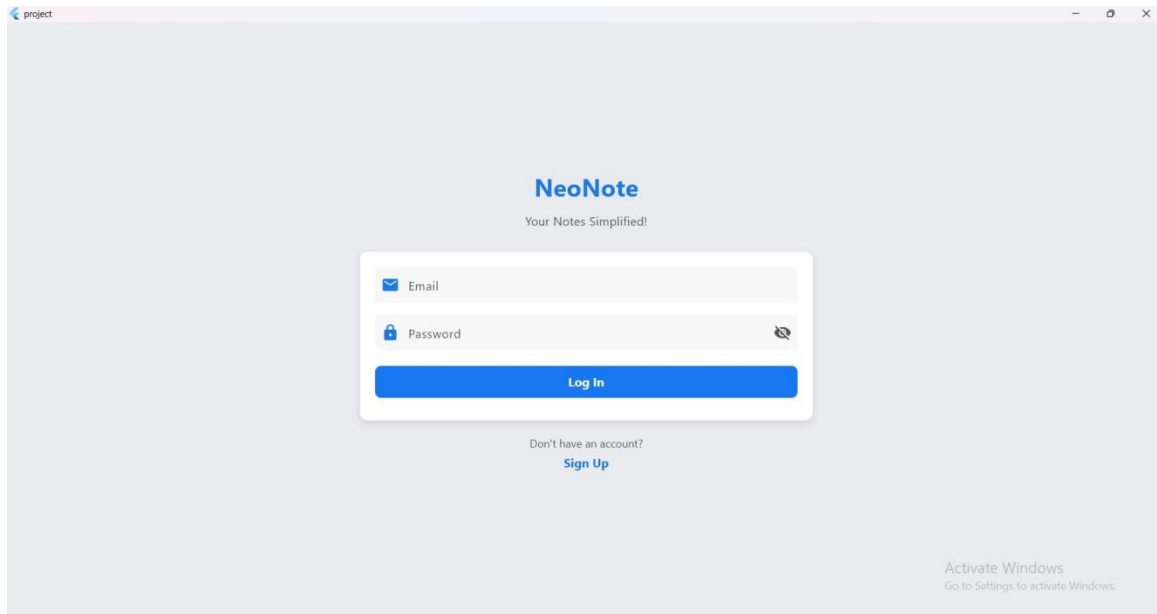
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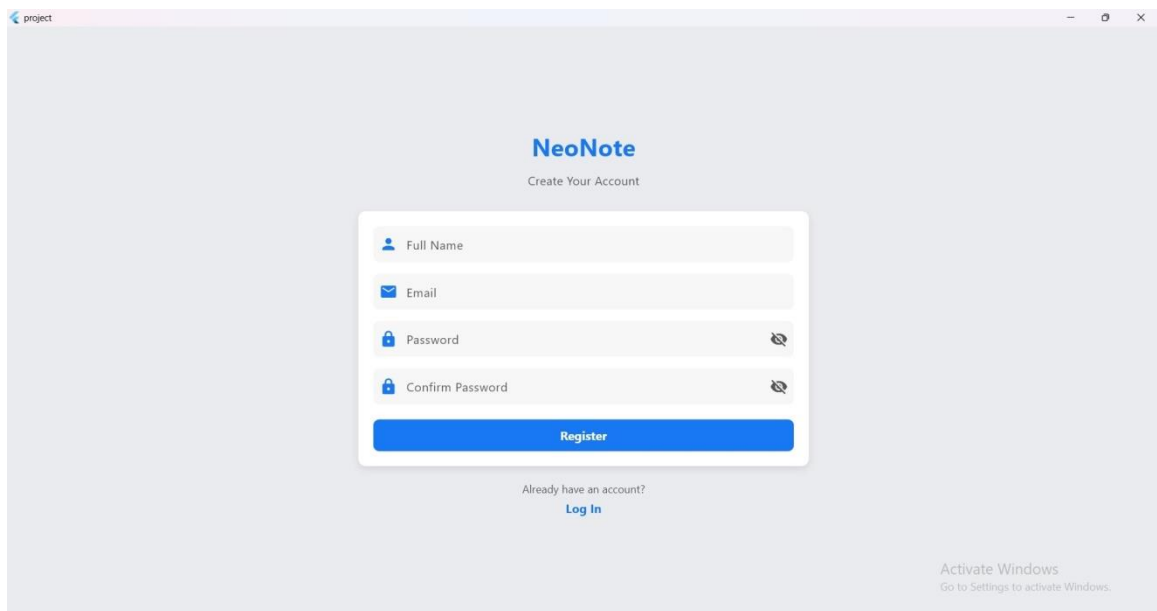
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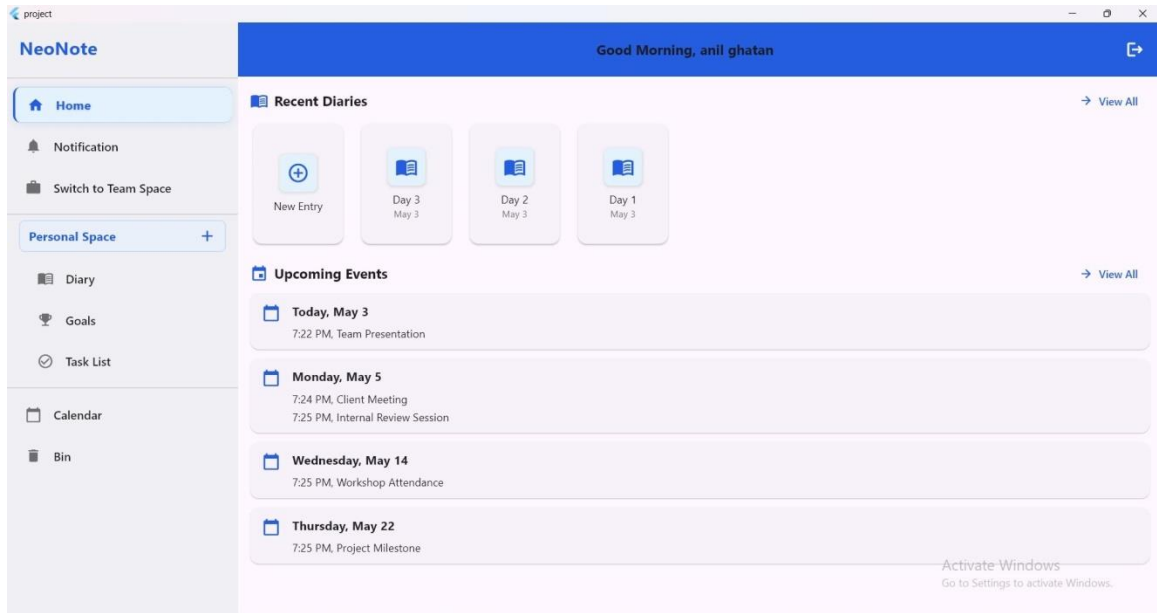
APPENDICES



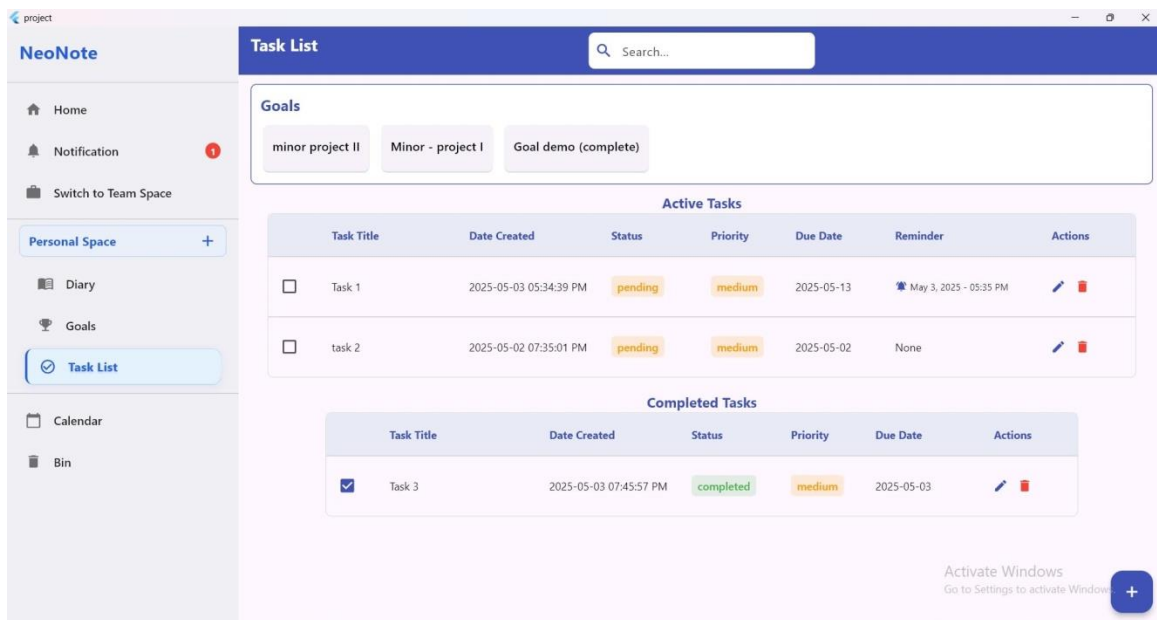
APPENDIX 1: LOGIN PAGE



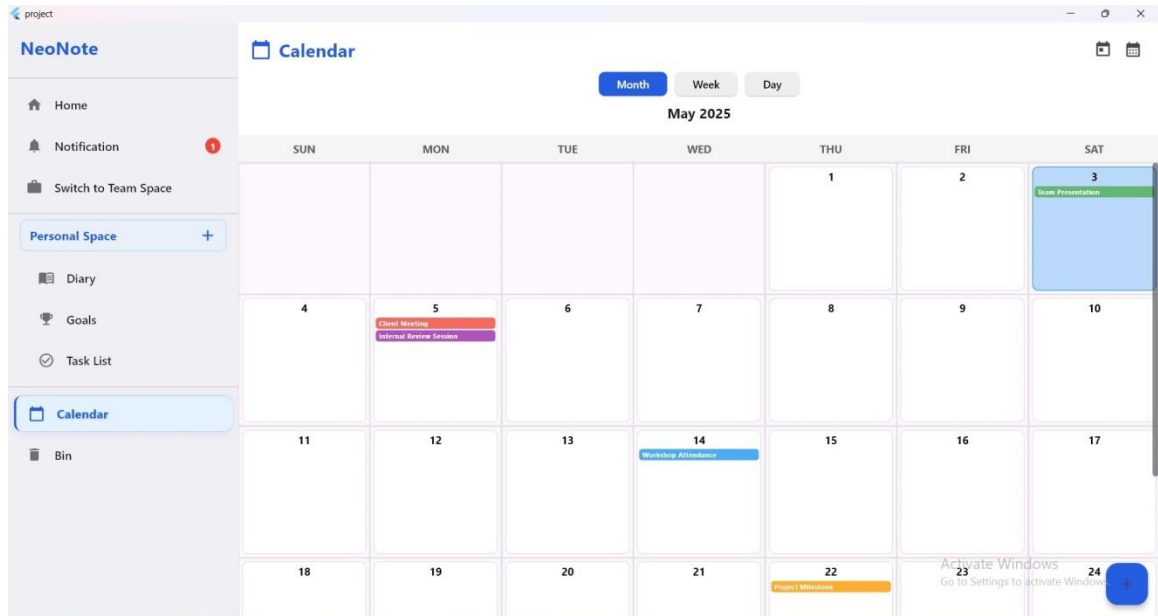
APPENDIX 2: REGISTER PAGE



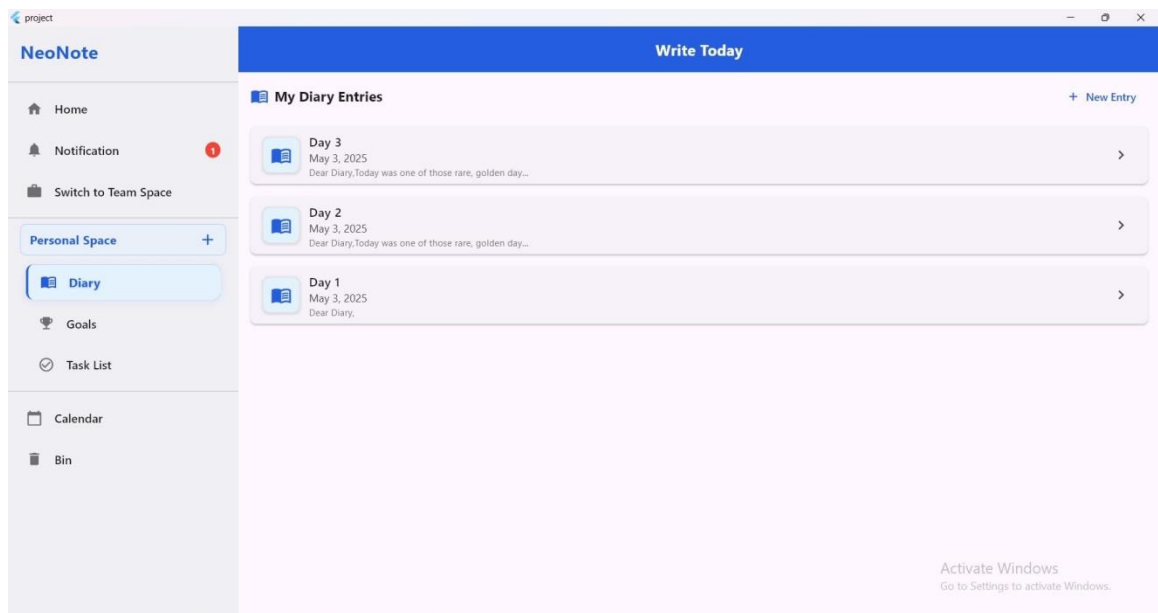
APPENDIX 3: PERSONAL SPACE DASHBOARD



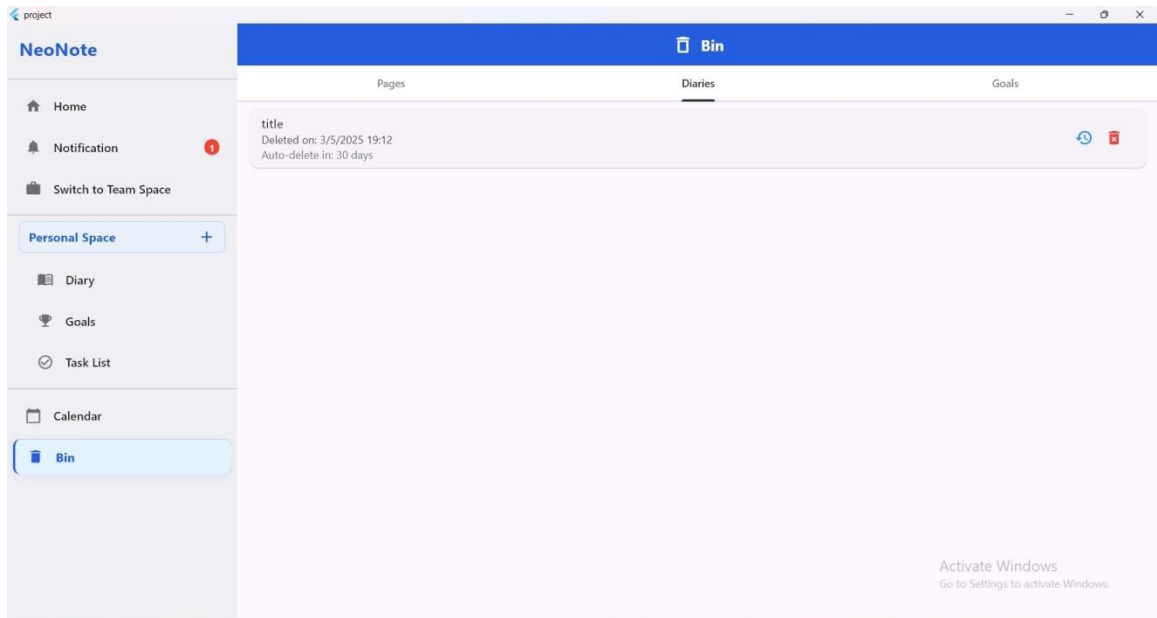
APPENDIX 4: TASK_LIST PAGE



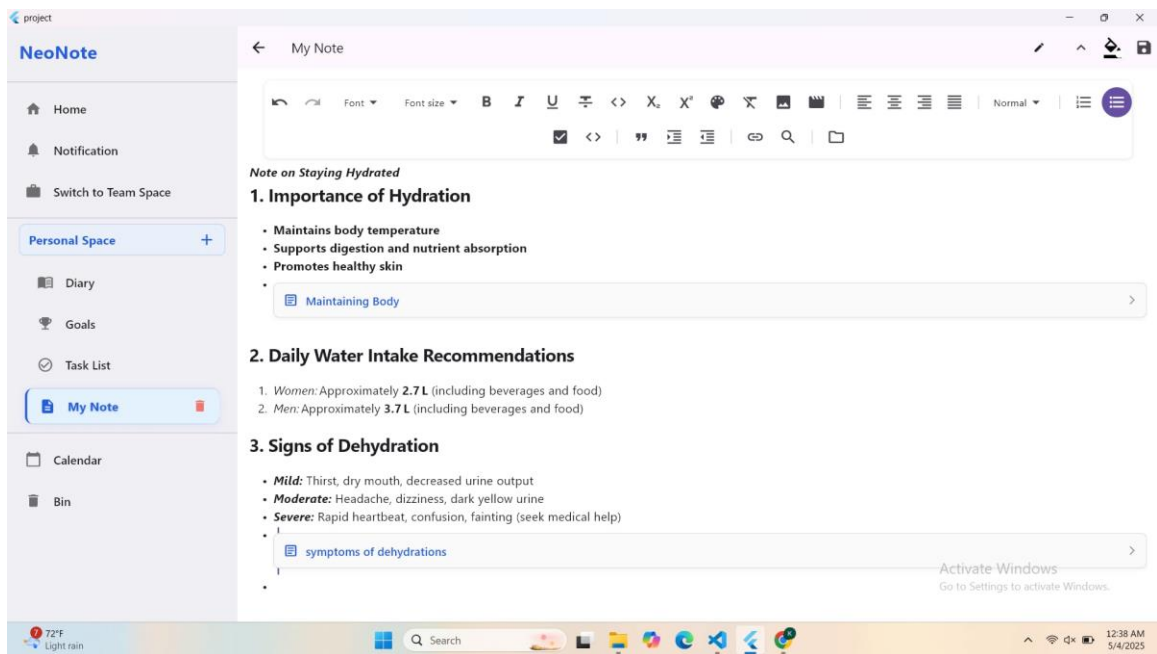
APPENDIX 5: CALENDER PAGE



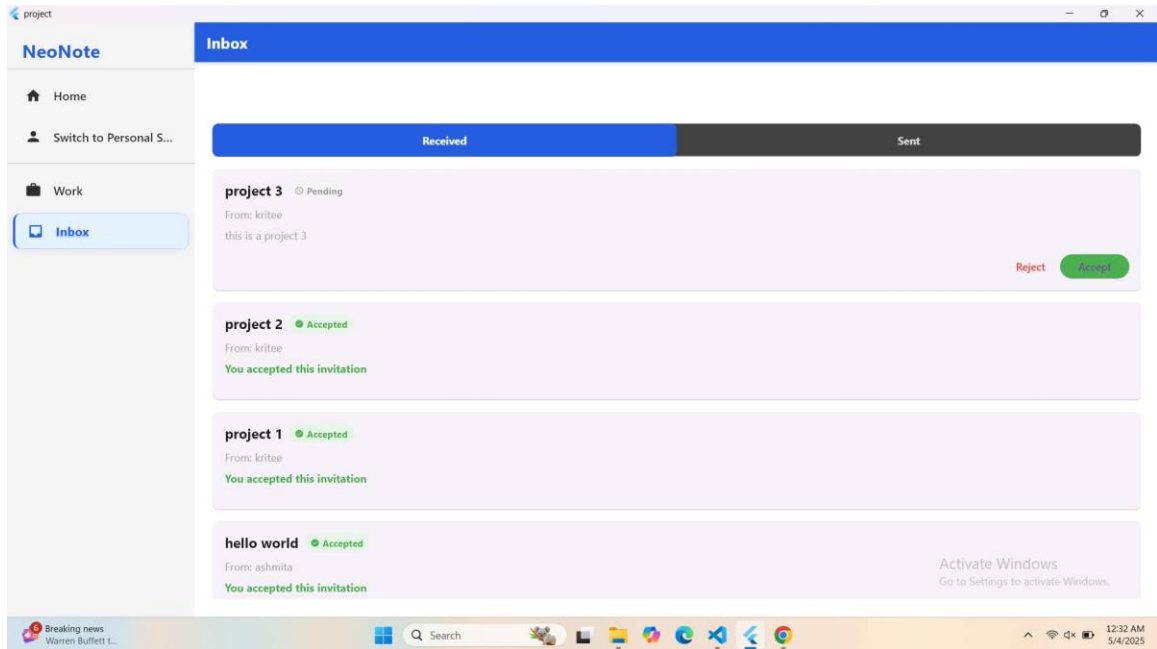
APPENDIX 6: DIARY PAGE



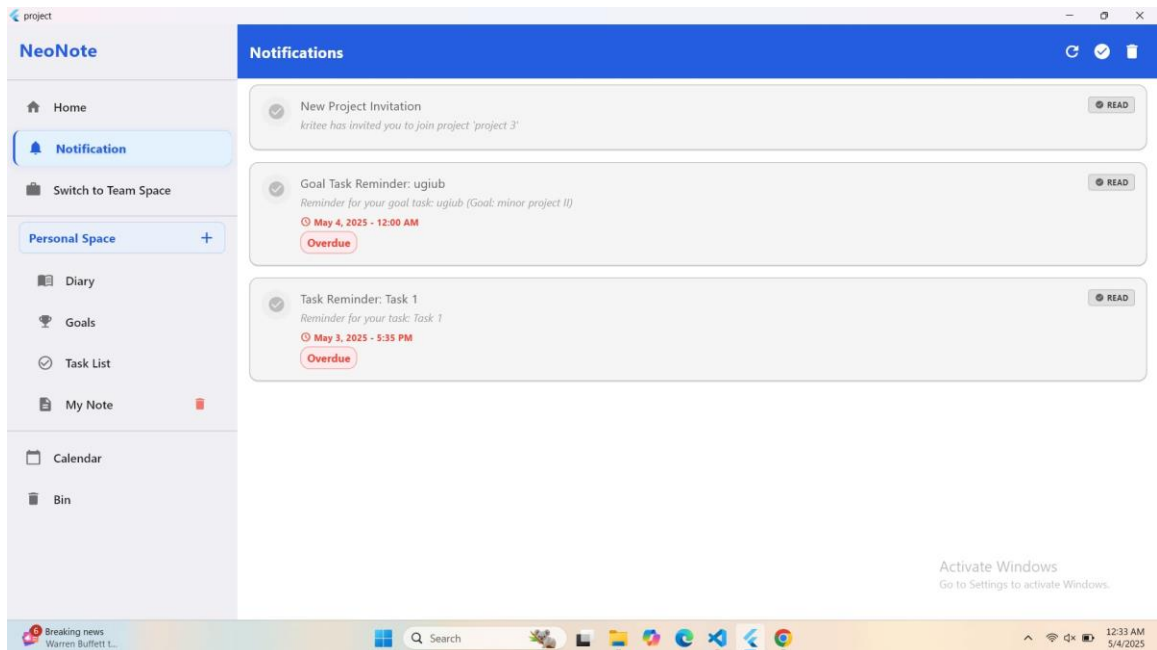
APPENDIX 7: BIN PAGE



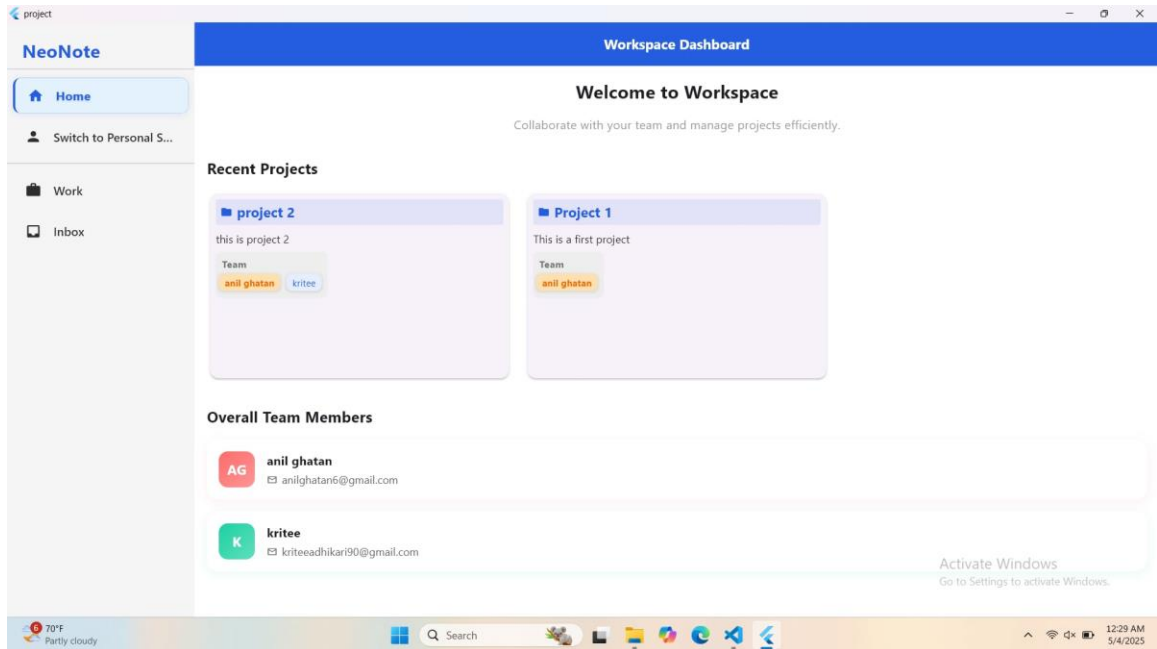
APPENDIX 8: NOTE PAGE



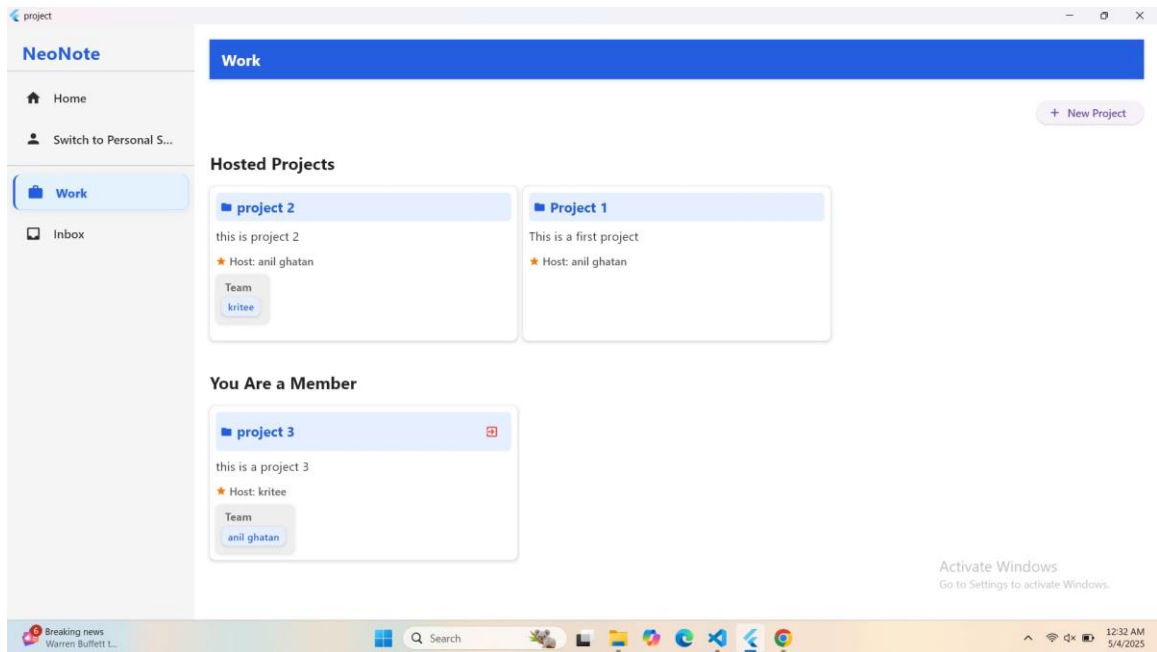
APPENDIX 9: NOTIFICATION PAGE1



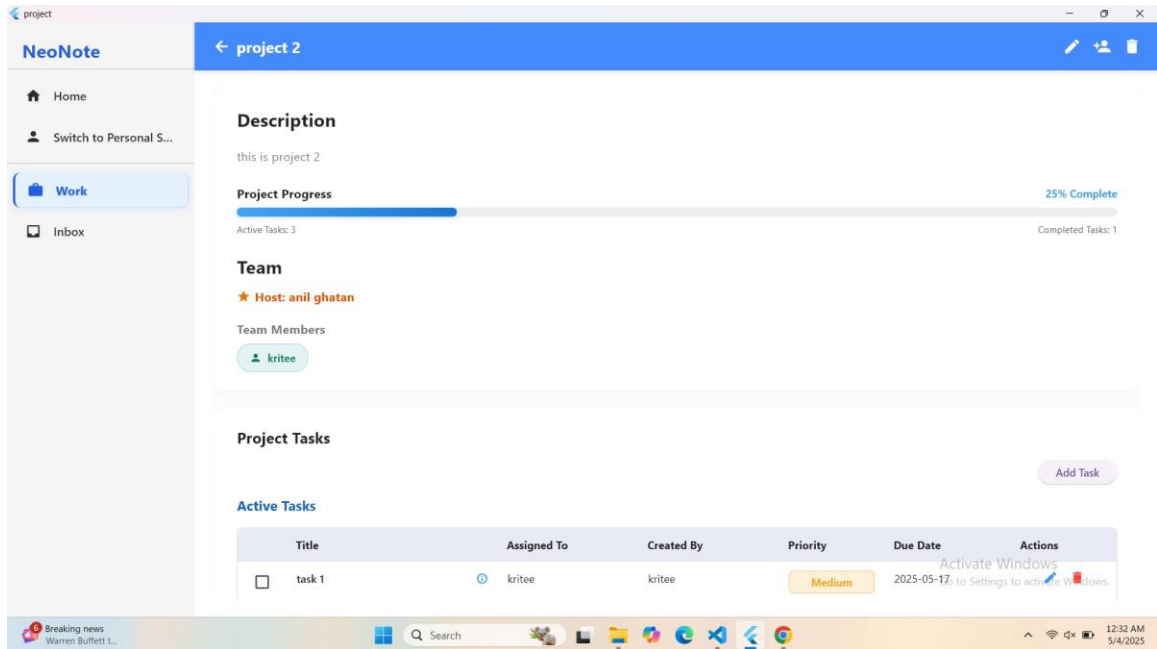
APPENDIX 10: NOTIFICATION PAGE2



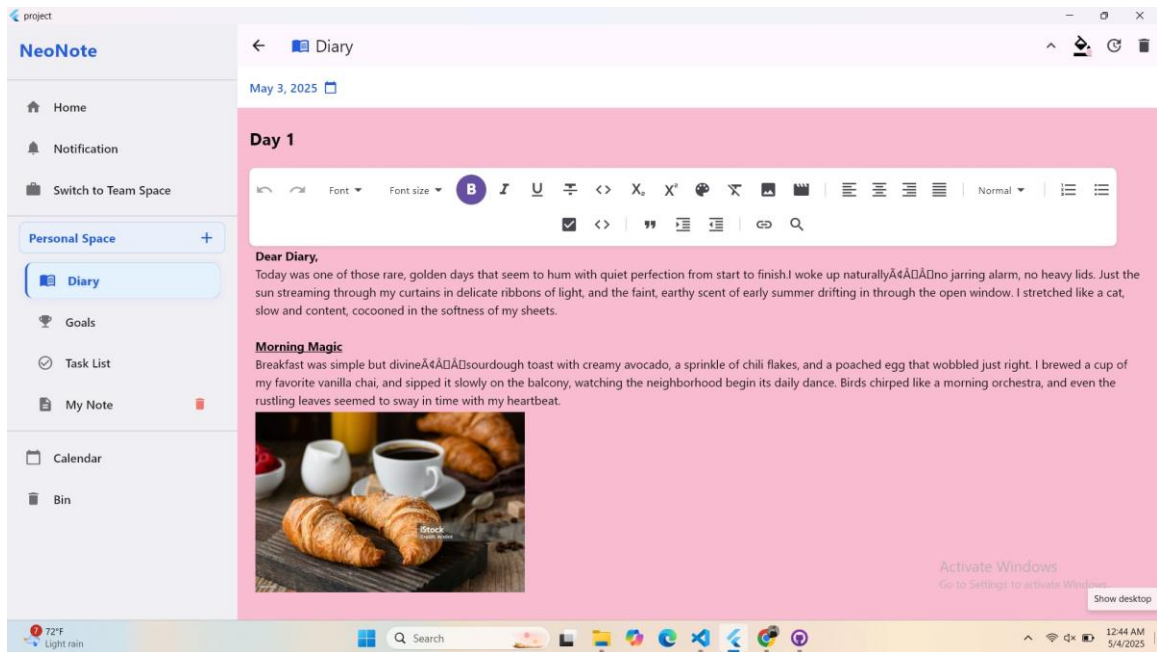
APPENDIX 11: WORKSPACE DASHBOARD



APPENDIX 12: WORK PAGE



APPENDIX 13: PROJECT_ADDING PAGE



APPENDIX 14: DIARY_WRITING PAGE